

HHS3190M
HUMAN PHYSIOLOGY AND FUNCTIONAL ANATOMY
FOR REHABILITATION SERVICES

HHS3190MJ
復康服務之人體生理學及功能性解剖學

Physiology L3: Membrane Transport
生理學 L3: 跨膜運輸

(Sem 1 AY 26-27 26-27學年第1學期)

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 - Osmosis 滲透
- Active Membrane Transport 主動膜運輸
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Revision: Structure and Properties of Cell Membrane

重溫：細胞膜的結構和特性

- **Cell membrane** 細胞膜
 - **Phospholipid bilayer** 雙層磷脂 + **Membrane proteins** 膜蛋白
- **Functions** 功能：
 - **Separate intracellular region and extracellular region**
分隔細胞內部和細胞外部
 - **Control substances entering and exiting the cell** 控制物質進出細胞
- **Property** 特性：
 - **Selective permeability** 選擇性滲透性：
 - Cell membrane allows some substances to pass in different rates while it restricts certain other substances to pass
細胞膜允許某些物質以不同的速率通過，同時限制某些其他物質通過

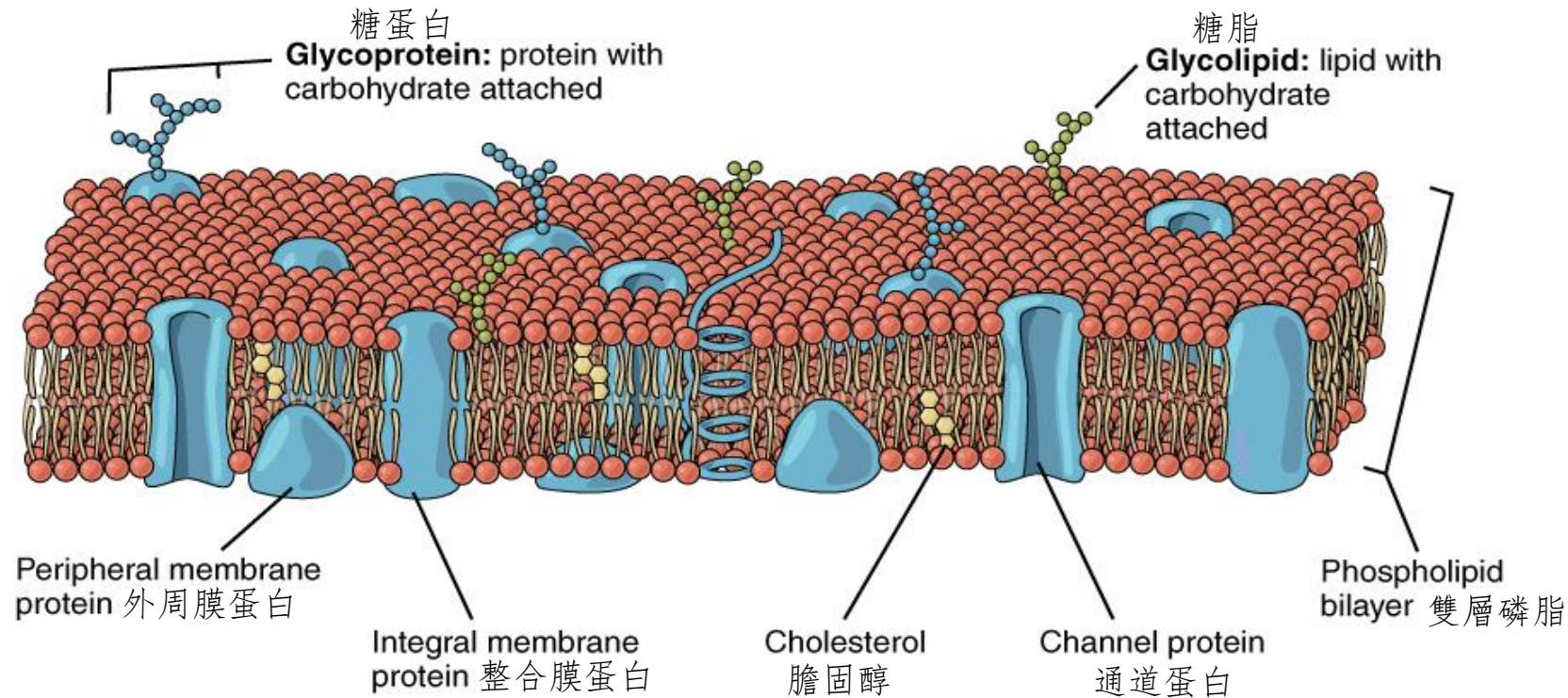
Selective permeability of cell membrane 細胞膜的**選擇性滲透性**:

(1) Small non-polar molecules pass the phospholipid bilayer quickly **非極性小分子**快速通過雙層磷脂

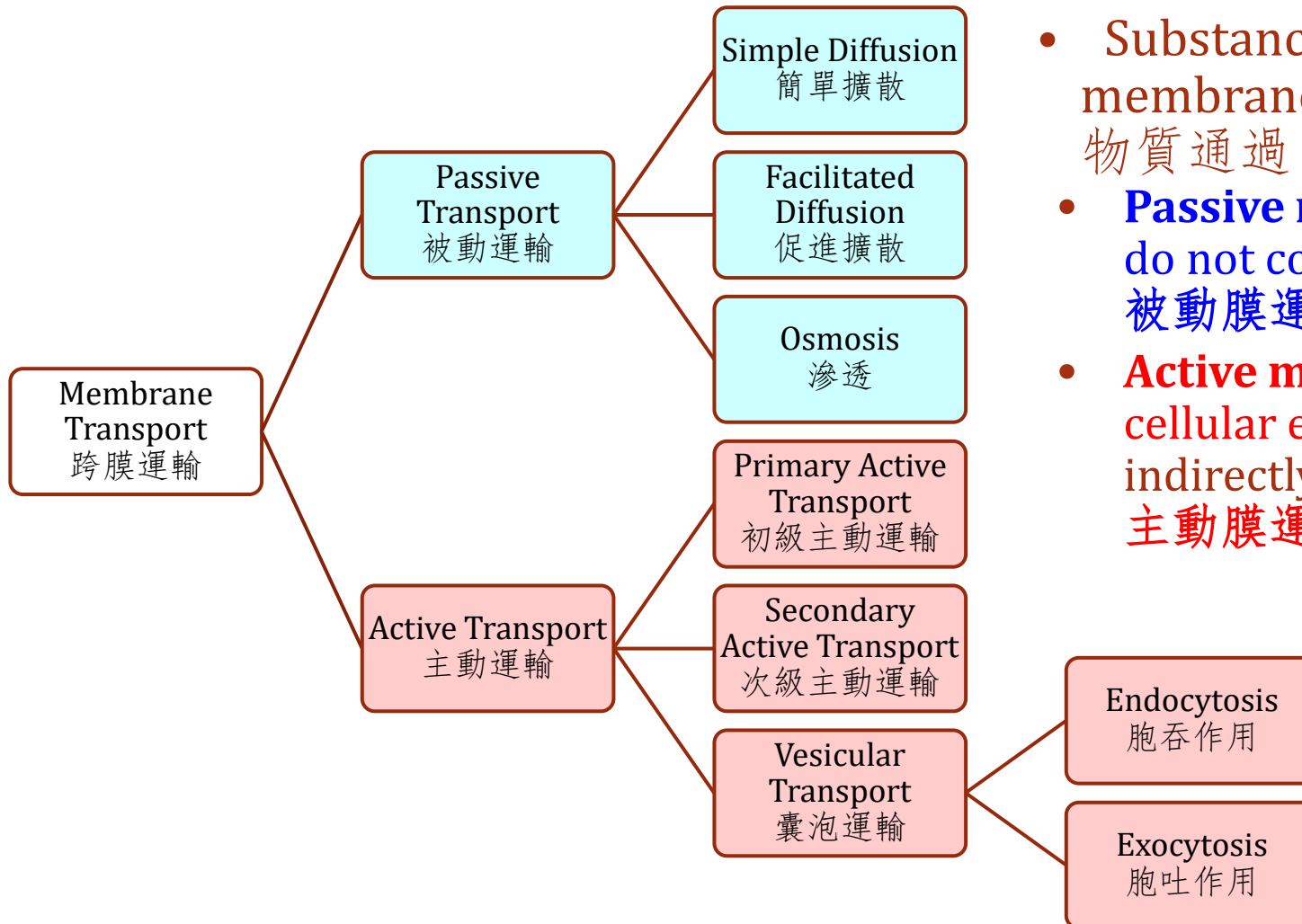
(2) Polar molecules or ions need to use certain transporter proteins (channel proteins or carrier proteins) to pass cell membrane

極性分子或離子需要使用某些轉運蛋白(通道蛋白或載體蛋白)通過細胞膜

(3) Big molecules (or large amount of substance at the same time) need to form vesicles to pass cell membrane **大分子**(或同時大量物質)需要形成囊泡通過細胞膜



Membrane Transport 跨膜運輸

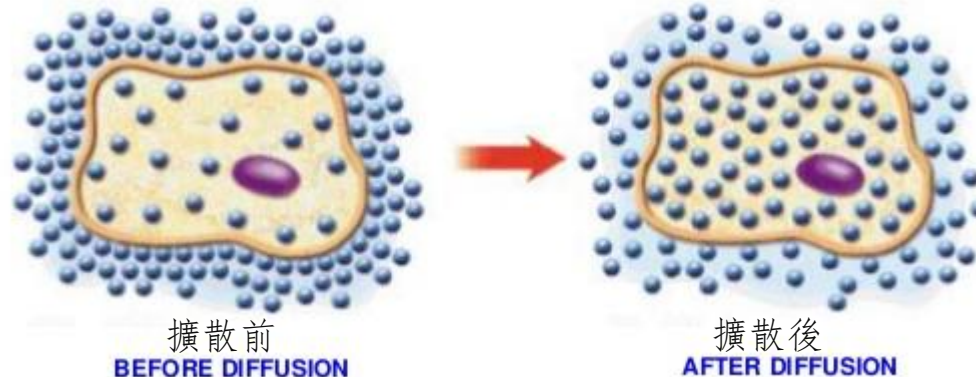


- Substances move through the cell membrane by 2 types of methods: 物質通過 2 種方法穿過細胞膜:
- **Passive membrane transport:** do not consume cellular energy
被動膜運輸: 無需消耗細胞能量
- **Active membrane transport:** cellular energy is consumed directly or indirectly
主動膜運輸: 細胞能量直接或間接消耗

Passive Membrane Transport

被動膜運輸

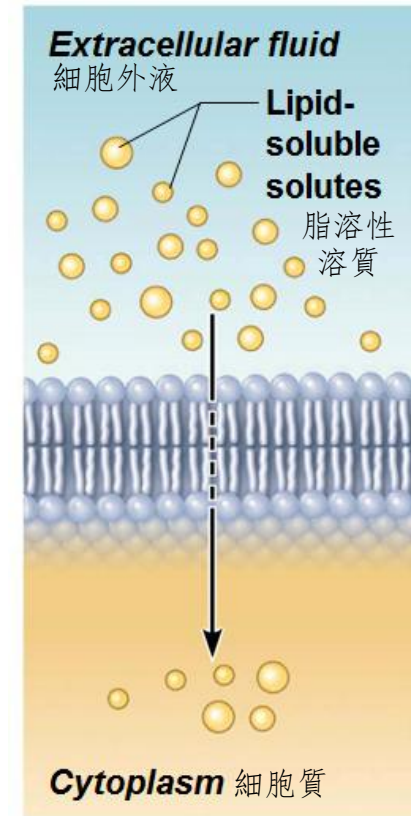
- Passive transport **does not require energy**
被動運輸不需要能量
- “**Diffusion 擴散**” :
- **Substance** moves from **region of higher concentration** to **region of lower concentration**
物質從高濃度區域移動到低濃度區域
- **Difference of concentrations (Concentration gradient) determines the rate of diffusion!**
濃度差異(濃度梯度)決定了擴散速率
- Steeper concentration gradient, Higher diffusion rate 更陡峭的濃度梯導致更高的擴散速率



Passive Membrane Transport

被動膜運輸

- **Simple diffusion** 簡單擴散
- **Nonpolar lipid-soluble substances** diffuse directly through **phospholipid bilayer**
非極性脂溶性物質直接通過雙層磷脂擴散
- Examples 例子:
 - Oxygen (O_2) 氧氣,
 - carbon dioxide (CO_2) 二氧化碳
 - fat-soluble vitamins 脂溶性維生素
 - Ethanol 乙醇



(a) Simple diffusion
of fat-soluble molecules directly through the phospholipid bilayer
脂溶性分子的簡單擴散直接通過雙層磷脂

Passive Membrane Transport

被動膜運輸

- **Facilitated diffusion 促進擴散**

- Certain **polar water-soluble molecules** (e.g. **glucose, amino acids, and ions**) are transported passively down their concentration gradient by:

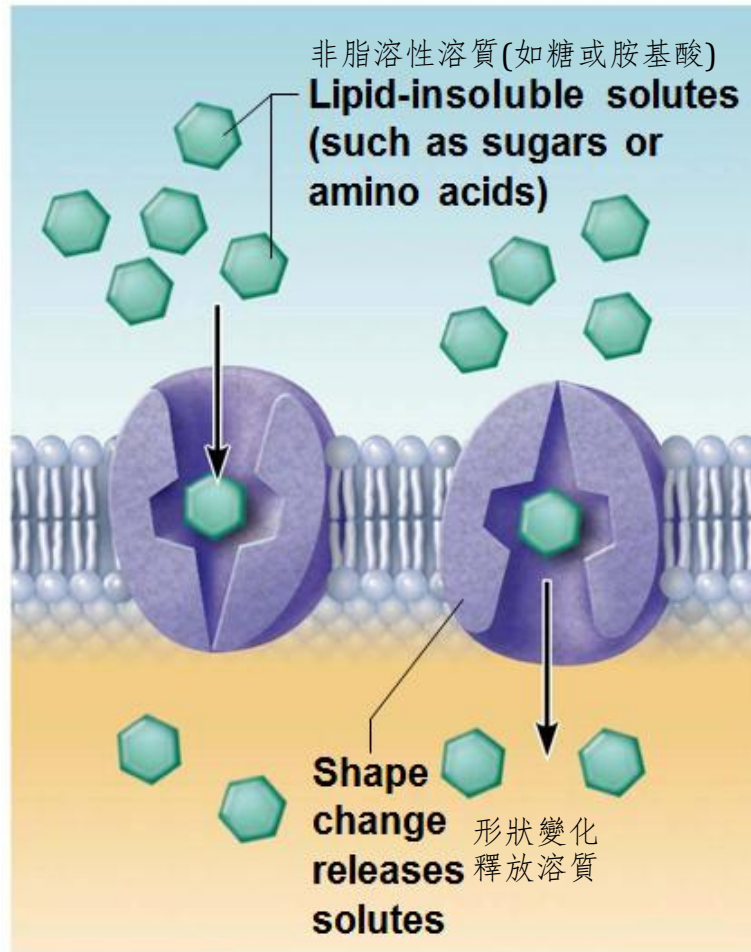
某些**極性水溶性分子**(例如**葡萄糖、氨基酸和離子**)通過以下方式被動地沿其濃度梯度向下運輸:

- **Carrier-mediated facilitated diffusion 載體介導的促進擴散**

- Substances bind to **carrier proteins** 物質與**載體蛋白**結合
- E.g. glucose, amino acids 例如葡萄糖、氨基酸

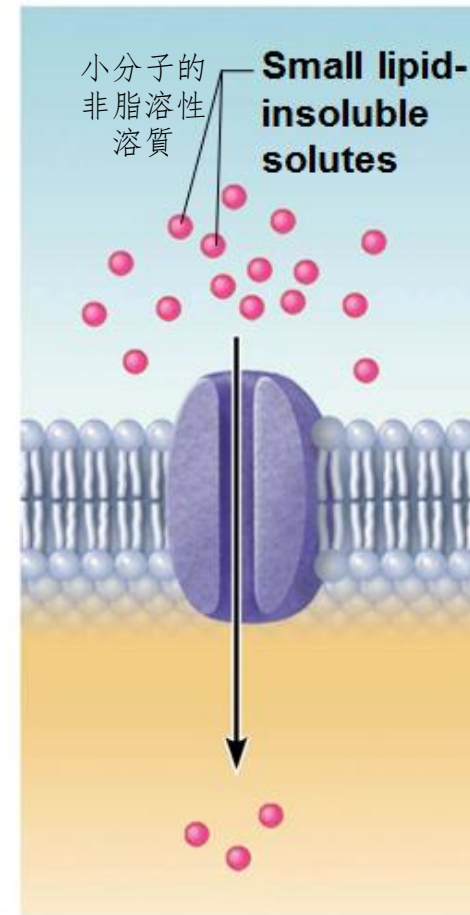
- **Channel-mediated facilitated diffusion 通道介導的促進擴散**

- Substances move through **channel proteins** 物質通過**通道蛋白**移動
- E.g. calcium ions, sodium ions, potassium ions 例如鈣離子、鈉離子、鉀離子



(b) Carrier-mediated facilitated diffusion via carrier protein specific for one chemical; binding of substrate causes carrier protein to change shape

載體介導的促進擴散
通過單一化學物質特異性
的載體蛋白進行;
底物的結合導致載體蛋白
改變形狀



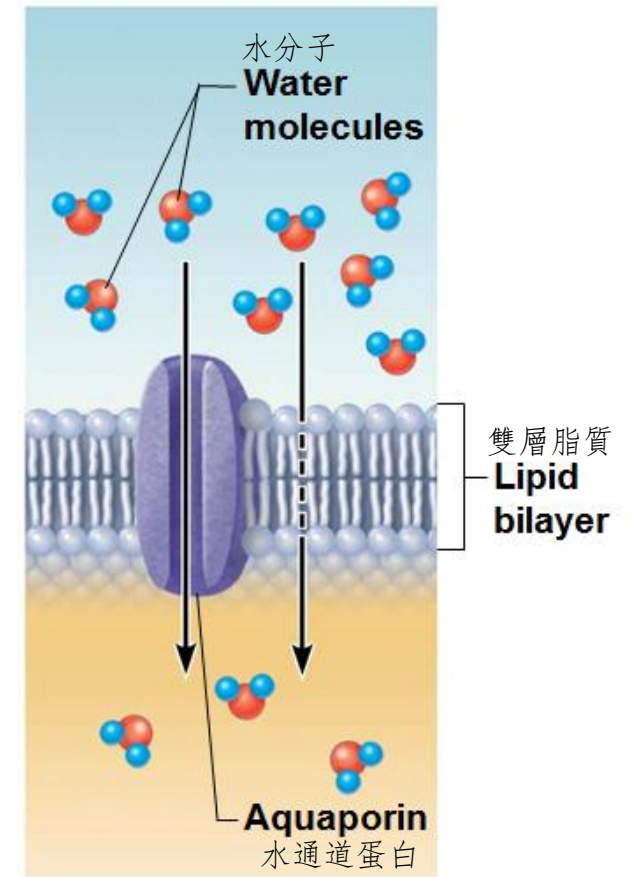
(c) Channel-mediated facilitated diffusion through a channel protein; mostly ions selected on basis of size and charge

通道介導的促進擴散
通過通道蛋白進行;
大多數是根據大小和
電荷選擇的離子

Passive Membrane Transport

被動膜運輸

- **Osmosis** 滲透作用
 - Movement of **water molecules** across a selectively permeable membrane from region of **higher water potential** to region of **lower water potential**
水分子穿過選擇性滲透膜
從高水勢區域移動到低水勢區域
- Water diffuses through cell membranes
水擴散通過細胞膜
 - Through lipid bilayer 通過雙層脂質
 - Through specific channel proteins called **aquaporins** 通過稱為水通道蛋白的特定通道蛋白



(d) Osmosis, diffusion of a solvent such as water through a specific channel protein (aquaporin) or through the lipid bilayer 滲透，溶劑(如水)通過特定通道蛋白(水通道蛋白)或雙層脂質擴散

- **Tonicity 滲透張力**

- Ability of a solution to change the shape of cells by altering the cells' internal water volume

溶液通過改變細胞內部水量來改變細胞形狀的能力

- **Isotonic solution** has same tonicity as inside the cell, so volume of cell remains unchanged

等滲溶液與細胞內具有相同的張力，因此細胞體積保持不變

- **Hypertonic solution** has higher osmolarity than inside cell, so water flows out of cell, resulting in cell shrinking

高滲溶液的滲透壓高於細胞內，因此水流出細胞，導致細胞收縮

- **Hypotonic solution** has lower osmolarity than inside cell, so water flows into cell, resulting in cell swelling or bursting

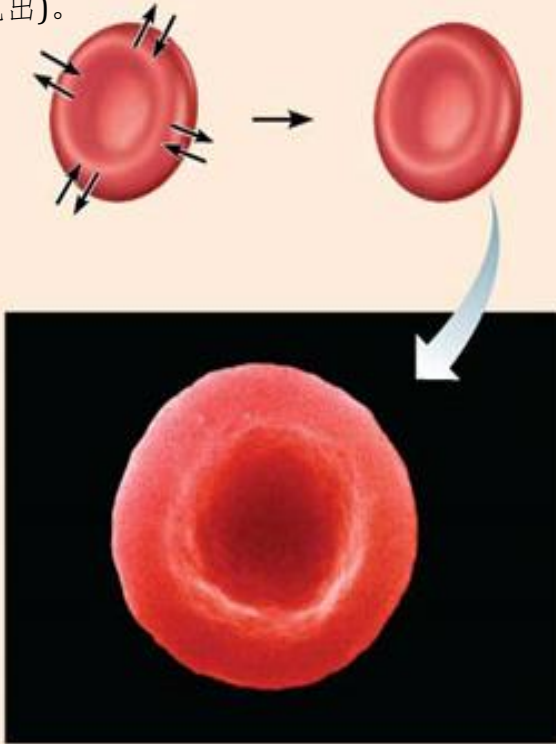
低滲溶液的滲透壓低於細胞內，因此水流入細胞，導致細胞膨脹或破裂

等滲溶液

(a) Isotonic solutions

Cells retain their normal size and shape in isotonic solutions (same solute/water concentration as inside cells; water moves in and out).

細胞在等滲溶液中保持其正常大小和形狀(溶質/水濃度與細胞內相同;水同時進入和流出)。

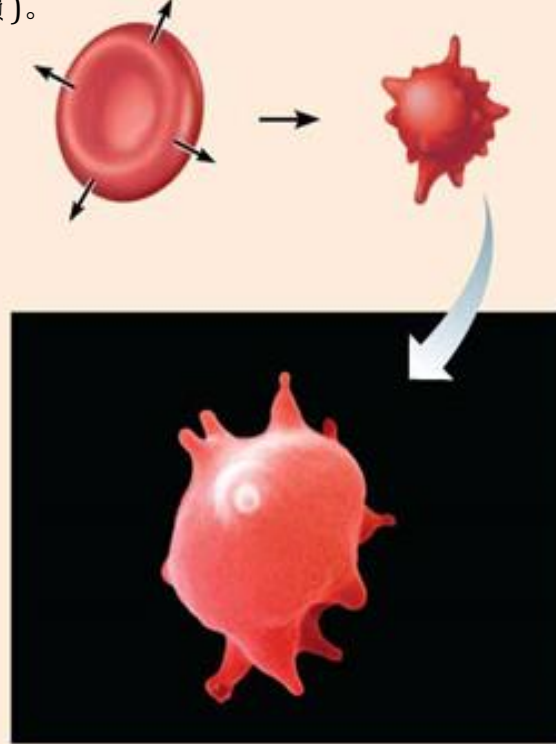


高滲溶液

(b) Hypertonic solutions

Cells lose water by osmosis and shrink in a hypertonic solution (contains a higher concentration of nonpenetrating solutes than are present inside the cells).

細胞通過滲透失去水分並在高滲溶液中收縮(含有比細胞內部存在的更高濃度的非滲透溶質)。

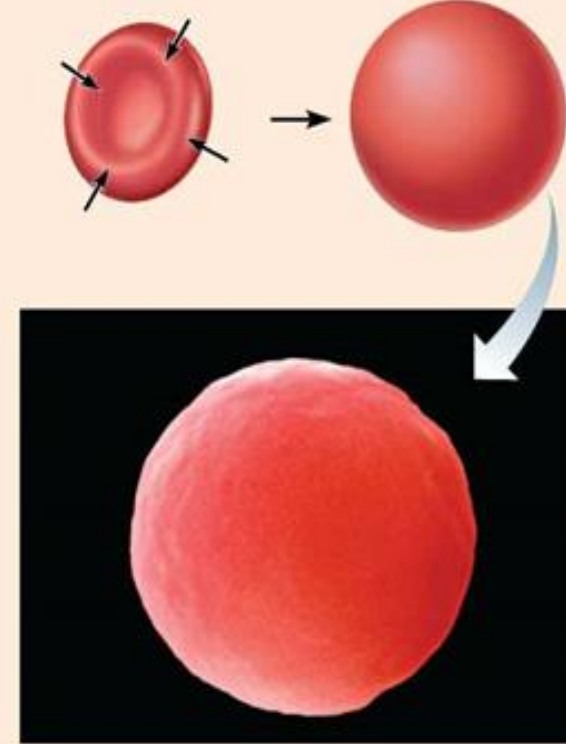


低滲溶液

(c) Hypotonic solutions

Cells take on water by osmosis until they become bloated and burst (lyse) in a hypotonic solution (contains a lower concentration of nonpenetrating solutes than are present inside cells).

細胞通過滲透吸收水分，直到它們在低滲溶液中膨脹和爆裂(裂解)(含有比細胞內部存在的濃度更低的非滲透性溶質)。



Active Membrane Transport

主動膜運輸

- Two major types of active membrane transport methods
兩種主要類型的主動膜運輸方法
 - Active transport 主動運輸
 - Vesicular transport 囊泡運輸/膜泡運輸
- Both **require cellular energy** to move molecules across plasma membrane for any of these reasons:
兩者都**需要細胞能量**來移動分子穿過細胞膜，原因如下：
 - Molecule is too large for channel proteins 分子對於通道蛋白來說太大
 - Molecule is not lipid soluble 分子不溶於脂質
 - Molecule is not able to move down concentration gradient
分子無法沿濃度梯度向下移動

Active Membrane Transport

主動膜運輸

- Active Transport 主動運輸
 - Requires special **carrier proteins (pumps)** 需要特殊的**載體蛋白(泵)**
 - Moves molecules **against their concentration gradient** (from low to high)
逆濃度梯度移動分子(從低到高)
 - This **requires energy directly or indirectly** 直接或間接地需要能量
 - Some pumps transport more than one substance 有些泵輸送多種物質
 - **Antiporters** 逆向轉運蛋白
 - transport one substance into cell while transporting a different substance out of cell
將一種物質運輸到細胞中，同時將另一種物質運輸出細胞
 - **Symporters** 同向轉運蛋白
 - transport two different substances in the same direction
沿同一方向輸送兩種不同的物質

Active Membrane Transport

主動膜運輸

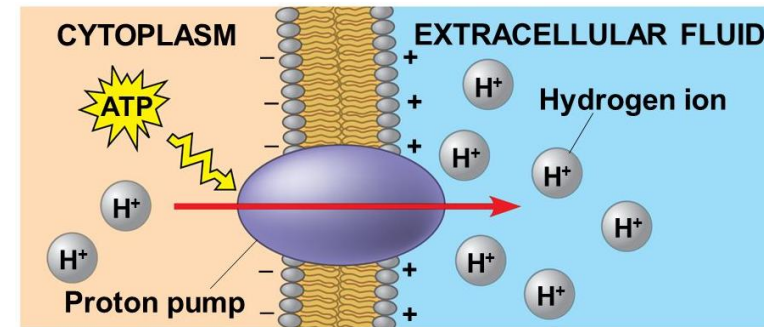
- Two types of active transport:
兩種類型的主動運輸：

- **Primary active transport**
初級主動運輸

- Required energy directly 直接需要能量

- **Secondary active transport**
次級主動運輸

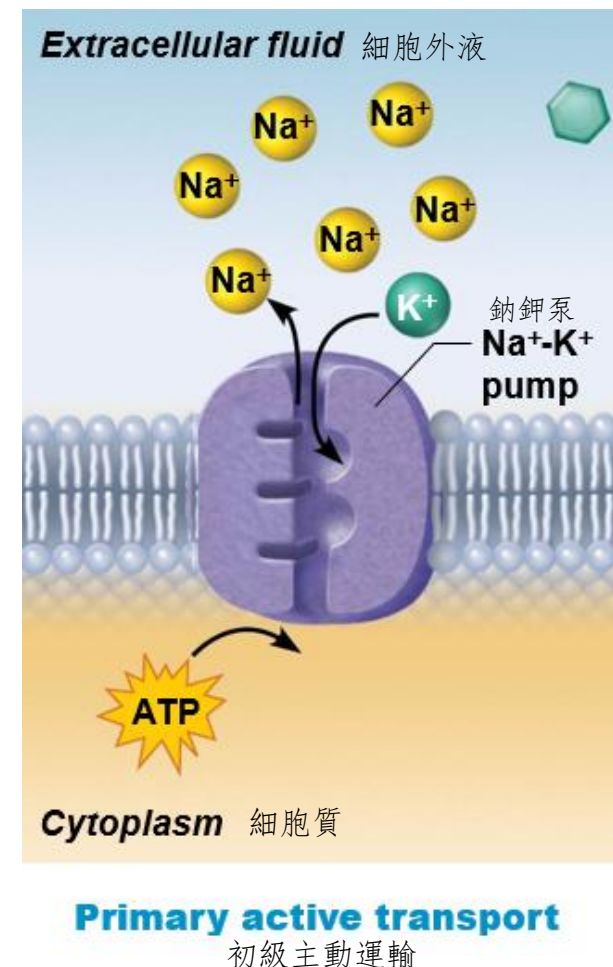
- Required energy indirectly from ionic gradients created by primary active transport
需要來自初級主動運輸消耗能量產生的離子梯度，間接需要能量



Active Membrane Transport

主動膜運輸

- Important carrier protein in human cells
人體細胞內重要的載體蛋白
- **Sodium-potassium pump (Na^+/K^+ ATPase)**
鈉鉀泵 (鈉鉀ATP酶)
 - Most studied pump
研究最多的泵
 - **Pumps 3 Na^+ out of cell and 2 K^+ back into cell**
將 3 個 Na^+ 泵出細胞，將 2 個 K^+ 泵入細胞
 - Present in all plasma membranes, but especially active in nerve cells and muscle cells
存在於所有細胞膜中，
但在神經細胞和肌肉細胞中尤其活躍



Active Membrane Transport

主動膜運輸

- **Vesicular Transport (Bulk Transport)** 囊泡運輸(大量運輸)
 - Involves transport of many molecules, large molecules (E.g. large protein molecules), and fluids across membrane by forming **vesicles**
通過形成**囊泡**進行許多分子、大分子(例如大蛋白質分子)和液體跨膜的運輸
 - **Requires cellular energy** 需要細胞能量
 - Vesicular transport processes include: 囊泡運輸過程包括:
 - **Endocytosis**: transport into cell 胞吞作用: 轉運到細胞中
 - **Exocytosis**: transport out of cell 胞吐作用: 從細胞中轉運出來

Active Membrane Transport

主動膜運輸

- **Endocytosis** 胞吞作用

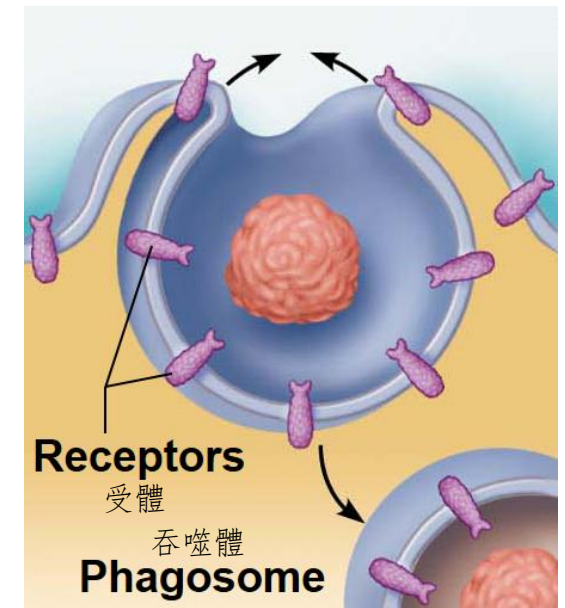
- Involves formation of **vesicles** by invagination of cell membrane to transport certain substance into cytoplasm
通過細胞膜內陷形成**囊泡**，以將某些物質運輸到細胞質中

- Example: **Phagocytosis**

例子：**吞噬作用**

- The cell engulfs a large particle by enclosing it within a vesicle called a phagosome. The phagosome is then combined with a lysosome.

細胞通過將大顆粒包裹在稱為吞噬體的囊泡中來吞噬它，然後將吞噬體與溶酶體結合。

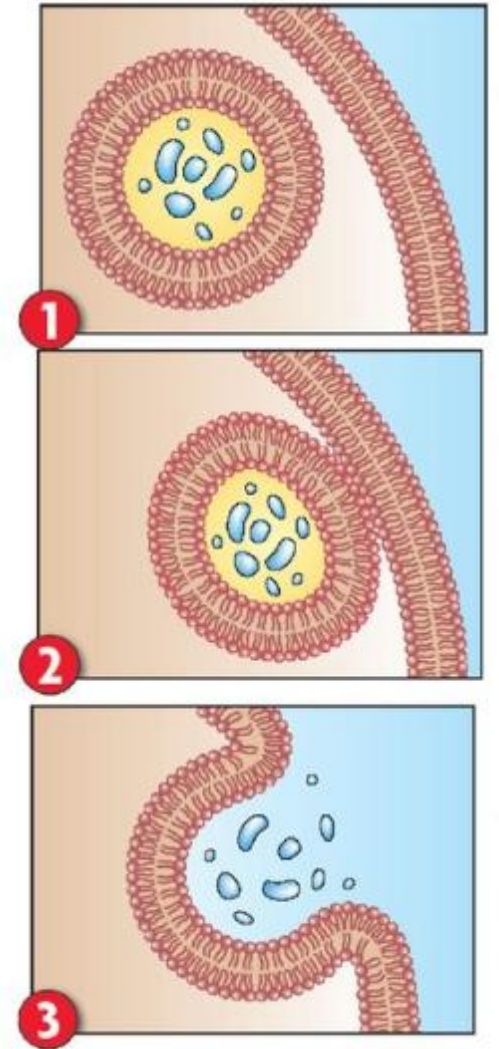


Active Membrane Transport

主動膜運輸

- **Exocytosis** 胞吐作用

- Process where material is ejected from cell by **secretory vesicle**
物質通過**分泌囊泡**從細胞中噴出的過程
- Secretory vesicle contains the molecules. The vesicle fuses with cell membrane and release the molecules to outside of cell.
分泌囊泡包含分子，囊泡與細胞膜融合並將分子釋放到細胞外。
- Examples 例子:
 - hormones, neurotransmitters, mucus, cellular wastes
激素、神經遞質、粘液、細胞廢物



Membrane Transport and Membrane Potential

跨膜運輸與膜電位

- Selective permeability of cell membrane leads to different concentrations of charged particles in cytoplasm and in extracellular fluid. This results in **membrane potential**, the difference of voltage across the cell membrane.
細胞膜的選擇性通透性導致細胞質和細胞外液中帶電粒子濃度不同，這導致**膜電位**，即細胞膜內外的電壓差。
- In resting state, **resting membrane potentials** of all human cells range from **-50mV to -100mV**, depending on cell type.
在靜息狀態下，所有人體細胞的**靜息膜電位**範圍為 **-50mV 至 -100mV**，具體取決於細胞類型。
- negative sign (-) indicates **inside of cell is more negative** than outside of cell
負號(-)表示**細胞內部**比細胞外部電壓更負

Membrane Transport and Membrane Potential

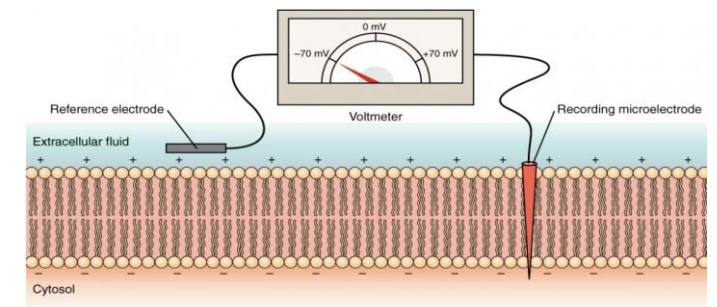
跨膜運輸與膜電位

- Generation and Maintaining of Resting Membrane Potential

產生和維持靜息膜電位

- (1) **Proteins with negative charge** are too large and polar. They cannot pass across cell membrane and will be trapped inside cell.
帶負電荷的蛋白質分子太大且極性太強，不能穿過細胞膜，會被困在細胞內。
- (2) Cell membrane is somehow more permeable to **K⁺**. K⁺ are transported out of cells by facilitated diffusion.
細胞膜某程度上對**鉀離子**的滲透性更強，鉀離子通過促進擴散從細胞中轉運出來。
- (3) **Action of Na⁺/K⁺ pump** brings more ions with positive charge out of cell.
鈉鉀泵的活動將更多帶正電荷的離子帶出細胞。

- Change of membrane potential is very important for the function of muscular system and nervous system.
膜電位的變化對肌肉系統和神經系統的功能非常重要。



L3 Revision Exercise L3 複習練習

- Screen the QR code below to complete all questions in this revision exercise.
掃描下面的二維碼以完成此複習練習中的所有問題。



References

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